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Resolving the Lagging Paradox in Financial Forecasting: An Explainable Heterogeneous Mixture-of-Experts Framework

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Abstract

Financial time-series forecasting remains difficult under non-stationarity and regime shifts, and models optimized for pointwise errors often exhibit conservative tracking and delayed reactions at turning points—the lagging paradox—weakening trading usefulness. We propose an interpretable hybrid forecasting framework for the CSI 300 index that decomposes linear and nonlinear dynamics: the linear component is modeled by ARIMA, while the nonlinear residual is learned by a semantically constrained heterogeneous Mixture-of-Experts (MoE) with structurally distinct experts targeting trend persistence, volatility clustering, and short-term reversal. A time-aware attention-based gating network adaptively routes and combines experts across regimes, gate-entropy regularization mitigates expert collapse, and a concept-bottleneck pathway maps residual patterns into a small set of human-interpretable concepts to improve transparency. On daily CSI 300 data (January 2022–May 2025) with a strict chronological split, the proposed model improves out-of-sample performance and achieves strong directional predictability; a long-only timing strategy based on predicted next-day returns delivers a Sharpe ratio of 2.86 with a maximum drawdown of -2.97%, outperforming the buy-and-hold benchmark. We further evaluate robustness under non-stationarity using rolling-window cross-validation over multiple market cycles, and extend the pipeline with execution-level safeguards for realistic trading frictions; in a 5-minute setting with transaction costs, the strategy remains economically viable and is supported by nonparametric statistical validation. Finally, a cross-market test on Bitcoin suggests partially transferable, risk-mitigating structures beyond a single benchmark market.

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Chapter 1: Introduction

1.1 Background Theory

Time-series forecasting is central to quantitative finance but remains difficult under non-stationarity, regime shifts, and low signal-to-noise ratios (Lo, 2004; Sezer, Gudelek, & Ozbayoglu, 2020). Linear baselines such as ARIMA model structured dependence under approximate stationarity, yet their assumptions are often violated in markets with nonlinear dynamics (Hyndman & Athanasopoulos, 2018; Siami-Namini, Tavakoli, & Namin, 2018). Hybrid decomposition therefore separates linear components from nonlinear residuals to reduce single-model bias and improve robustness (Khashei & Bijari, 2011; Zhang, 2003), while deep sequence models (e.g., LSTM) enhance capacity for nonlinear temporal dependence (Fischer & Krauss, 2018). However, minimizing pointwise errors can induce lagging behavior and weaken trading relevance, motivating regime-aware specialization: MoE routes inputs to expert subnetworks via gating, and attention-based encoding further supports context-dependent routing (Fedus, Zoph, & Shazeer, 2022; Jacobs, Jordan, Nowlan, & Hinton, 1991; Lim, Arik, Loeff, & Pfister, 2021). In parallel, financial deployment requires auditability (Basel Committee on Banking Supervision, 2005), so explanations should be validated for faithfulness and stability rather than assumed (Adebayo et al., 2018; Fong & Vedaldi, 2017; Lundberg & Lee, 2017). Moreover, execution frictions are first-order in practice—especially at high frequency where turnover and costs can erase predictive edge (O'Hara, 2015; Perold, 1988)—which underscores the need for non-stationarity-aware evaluation and anti-snooping inference (Bergmeir & Ben'ıtez, 2012; Hansen, 2005; White, 2000).

1.2 Problem Statement

Despite progress in hybrid forecasting, several gaps still limit practical deployment in finance:

1. **Regulatory interpretability:** Neural components that improve flexibility (e.g., LSTM and MoE gating) are often opaque, yet deployment requires transparent and auditable reasoning (Basel Committee on Banking Supervision, 2005).

2. **Lagging paradox:** Pointwise loss minimization can encourage price-tracking forecasts that miss turning points, improving RMSE/MAE while weakening directional and trading relevance.

3. **Limited regime specialization:** Many MoE implementations use homogeneous experts, underexploiting heterogeneous market behaviors (trend, volatility, reversals) that call for distinct inductive biases (Fedus et al., 2022; Jacobs et al., 1991).

4. **Predictive–economic mismatch (high frequency):** Even with directional edge, naive execution can induce fat-tailed turnover and cost clustering that erode alpha under realistic frictions (O'Hara, 2015; Perold, 1988).

5. **Fragile empirical inference:** Fold-wise outcomes are often non-normal and repeated comparisons amplify data-snooping risk, motivating nonparametric tests and multiple-

comparison safeguards (Hansen, 2005; Shapiro & Wilk, 1965; White, 2000; Wilcoxon, 1945).

6. **Unverified explanations:** Post-hoc interpretability is frequently reported without falsifiable faithfulness and stability checks, limiting decision-support trust (Adebayo et al., 2018; Fong & Vedaldi, 2017).

Accordingly, this thesis asks whether an interpretable hybrid architecture can (i) improve forecasting and directional predictability while mitigating lagging, (ii) maintain stable regime-aware routing via heterogeneous experts, and (iii) deliver economically viable signals under trading frictions with credible statistical validation and cross-market generalization.

1.3 Research Contributions

To address these gaps, this thesis proposes an interpretable hybrid framework and makes the following contributions:

1. **Heterogeneous expert architecture:** We develop a heterogeneous MoE with structurally distinct experts aligned to market behaviors (trend, volatility patterns, and short-term reversals), improving inductive bias matching compared with homogeneous expert sets (Fedus et al., 2022; Jacobs et al., 1991).

2. **Transparency by design with falsifiable explanation checks:** We integrate a concept-bottleneck pathway for concept-level interpretability (Basel Committee on Banking Supervision, 2005; Koh et al., 2020), and we evaluate explanations using faithfulness and stability diagnostics (e.g., perturbation-based tests and sanity checks) to strengthen reliability (Adebayo et al., 2018; Fong & Vedaldi, 2017).

3. **Friction-aware evaluation and execution-level robustness:** Beyond forecasting metrics, we evaluate economic utility under long-only constraints and explicit transaction costs, and we propose an execution-level risk-control wrapper that filters low-edge and high-uncertainty trades to reduce over-trading and tail cost events in high-frequency settings (O'Hara, 2015; Perold, 1988).

Chapter 2: Literature Review

Financial time-series forecasting is difficult because markets are non-stationary and regime-dependent with a low signal-to-noise ratio, so models that look accurate in-sample often weaken under distribution shift (Lo, 2004; Sezer et al., 2020). Moreover, optimizing pointwise forecast errors can encourage conservative tracking (the "lagging" failure mode), reducing trading relevance even when standard accuracy metrics improve (Makridakis, Spiliotis, & Assimakopoulos, 2018). Classical linear baselines such as ARIMA remain useful for modeling structured dependence under approximate stationarity (Hyndman & Athanasopoulos, 2018), but purely linear assumptions are misaligned with nonlinear and switching dynamics, motivating hybrid decomposition that separates linear structure from nonlinear residual components (Khashei & Bijari, 2011; Zhang, 2003). Deep learning further expands capacity for nonlinear temporal patterns: LSTM-based models have shown competitive results in financial prediction tasks (Fischer & Krauss, 2018), and CNN-style modules can capture localized structures such as short-lived volatility bursts (Hoseinzade & Haratizadeh, 2019).

Because financial dynamics alternate across trend, volatility, and reversal regimes, regime-aware specialization becomes central: the Mixture-of-Experts (MoE) paradigm formalizes expert specialization and input-dependent routing via a gating mechanism (Jacobs et al., 1991), and modern sparse routing improves scalability by activating only a subset of experts per input (Fedus et al., 2022; Rajbhandari et al., 2022). Attention mechanisms further enable context-dependent weighting of historical information (Vaswani et al., 2017), and forecasting architectures such as Temporal Fusion Transformers illustrate how selective temporal attribution can support interpretability alongside performance (Lim et al., 2021).

In finance, interpretability is an operational constraint (Basel Committee on Banking Supervision, 2005); while post-hoc attribution is widely used (Lundberg & Lee, 2017), concerns about reliability in high-stakes settings motivate intrinsically interpretable designs such as concept bottlenecks and a broader responsible-XAI perspective (Koh et al., 2020; Rudin, 2019). Accordingly, explanation quality is increasingly validated empirically: perturbation-based faithfulness tests assess whether top-ranked features drive stronger degradation under deletion and faster recovery under insertion (Fong & Vedaldi, 2017; Petsiuk, Das, & Saenko, 2018), and sanity-check evidence shows explanations can be brittle under refits and randomization, motivating stability diagnostics (Adebayo et al., 2018).

These issues become sharper at high frequency, where microstructure effects inject noise and distort short-horizon predictability, and turnover and transaction costs can dominate realized performance (Almgren & Chriss, 2001; O'Hara, 2015). Finally, credible trading-oriented claims require evaluation and inference that respect non-stationarity and multiple testing: rolling-window protocols better reflect evolving regimes (Bergmeir & Ben'ıtez, 2012), and data-snooping corrections are essential when many folds or model candidates are compared (Hansen, 2005; White, 2000).

Together, this literature motivates our regime-aware MoE with time-aware gating, empirically validated explanation reliability, and friction-aware backtesting with statistically defensible inference.

Chapter 3: Methodology

The proposed study implements a hybrid forecasting framework that integrates linear statistical modeling with a semantically constrained deep learning architecture. The methodology is designed to address the "black box" nature of traditional deep learning while resolving the lagging issues common in financial forecasting.

3.1 The Hybrid ARIMA-LSTM Backbone

Following the decomposition principle proposed by Zhang (2003) and further validated by Khashei and Bijari (2011), we assume the financial time series y_t comprises a linear component L_t and a non-linear residual component N_t. The data generating process is defined as:

$$y_t = L_t + N_t + \varepsilon_t \tag{3.1}$$

where L_t denotes the linear trend captured by the statistical model, N_t represents the complex non-linear patterns, and ε_t is the random noise. The linear component is modeled using the Auto-Regressive Integrated Moving Average (ARIMA) method, while the non-linear residuals are passed to the deep learning component described in the subsequent sections.

3.2 Mixture of Experts with Semantic Partitions and Heterogeneous Architectures

Homogeneous base learners are usually used and result in the restriction of the standard ensemble methods in getting the variation of the inductive biases needed by the financial data. We will alleviate this drawback with a new Mixture-of-Experts (MoE) layer where every expert is optimized structurally to perform a certain semantic task.(Jacobs et al., 1991; Shi et al., 2024)
The input feature space is initially divided into three semantic clusters to exclude the interference of various market forces.(Hasbrouck, 2007) The Trend cluster combines the historical closing prices and moving averages to reflect long term direction, Volatility cluster uses rolling standard deviations and high-low spreads to reflect the market stress and Reversal cluster uses the RSI indicator and bias of moving averages to signal the mean-reversion. In contrast to the traditional models of MoE that deploy the same sub-networks, we deploy the work of a heterogeneous architecture of each semantic group.
To begin with, price trends have long-term time dependencies, and therefore the Trend Expert uses stacked Long Short-Term Memory (LSTM) network.(Yu, Si, Hu, & Zhang, 2019)This architecture has been selected to counter the vanishing gradient issue and identify long term market trends:

$$h_t^{(trend)} = \text{LSTM}(x_t^{(trend)}, h_{t-1}) \tag{3.2}$$

Secondly, volatility often manifests as abrupt, highly localized spikes or even flash crashes. Given that Convolutional Neural Networks (CNNs) excel at capturing local patterns (Hoseinzade & Haratizadeh, 2019), we feed volatility-specific features (e.g., high-low spreads and

rolling standard deviations) into the Volatility Expert, which employs a 1D convolutional layer to effectively extract these localized high-frequency stress patterns.

$$h^{(vol)} = \sigma(\mathbf{W}_{conv} * x^{(vol)} + b_{conv}) \tag{3.3}$$

Third, the mean reversion can be a simpler, state-dependent relationship. As a result, the Reversal Expert can be trained as a GAP-based lightweight Multilayer Perceptron (MLP):

$$h^{(rev)} = \text{ReLU}(\mathbf{W}_{mlp} \cdot \text{GAP}(x^{(rev)}) + b_{mlp}) \tag{3.4}$$

In order to efficiently manage such professionals, we present an Advanced Time-Aware Gating Mechanism. Common gating networks are based on the steady global context, which can miss out any temporary shifts in regimes(Lim et al., 2021). Rather, our gating network uses a specialized LSTM to encode the time dynamics of market states, which is used to produce hidden states H_{gate}. We use Self-Attention mechanism to dynamically weight the time steps of the history that are most relevant to the current regime: e.g. the past crash events:

$$A = \text{softmax}\left(\frac{H_{gate}H_{gate}^{\top}}{\sqrt{d_k}}\right)H_{gate} \tag{3.5}$$

Attention output A is fed through to come up with the final expert weight g_t. Moreover, to ensure that the gating network does not settle to a trivial solution, a phenomenon called Expert Collaps we introduce an Enforced Non-Collapsing mechanism(Fedus et al., 2022; Shazeer et al., 2017). This is done through a combination of the learned gating distribution $g_{learned}$ learned and a uniform prior u:

$$g = (1-\beta)g_{learned} + \beta \cdot u \tag{3.6}$$

where the exploration noise strength is controlled by an amount of β. This will make the model diverse in terms of the utilization of experts(Rajbhandari et al., 2022).

3.3 Model Optimization and Implementation Details

To meet regulatory requirements for transparency (Basel Committee on Banking Supervision, 2005), the framework incorporates interpretability directly into the learning process. We introduce a Concept Bottleneck layer (Koh et al., 2020) parallel to the MoE branch. The context vector is projected into a low-dimensional concept space $z_{\text{concept}} = W_c c + b_c$. To ensure that these concepts are distinct and not redundant, we impose a Decorrelation Regularization $\mathscr{L}_{\text{decor}} = \sum_{i \neq j} C_{ij}^2$, where C is the covariance matrix of the concept batch. This constraint forces the model to learn orthogonal, independent semantic features.
The final prediction is synthesized via an α-Fusion mechanism. An adaptive gate $\alpha = \sigma(w_{\alpha}^{\top} c + b_{\alpha})$ combines the complex non-linear prediction from the MoE branch (r_{expert}) and the interpretable linear prediction from the concept branch (r_{concept}):

$$r_{\text{final}} = \alpha \cdot r_{\text{expert}} + (1-\alpha) \cdot r_{\text{concept}} \tag{3.7}$$

The total objective function minimizes prediction error while maximizing gate entropy and concept separability:

$$\mathscr{L}_{\text{total}} = \mathscr{L}_{\text{MSE}} + \lambda_{\text{ent}}\mathscr{L}_{\text{gate}} + \lambda_{\text{decor}}\mathscr{L}_{\text{decor}} \tag{3.8}$$

Prior to model training, strict data preprocessing is applied. Consistent with Zhang (2003), the Augmented Dickey-Fuller (ADF) test (Dickey & Fuller, 1979) is used to ensure stationarity for the ARIMA component, with non-stationary series transformed via differencing ($y'_t = y_t - y_{t-1}$). For the neural network components, Min-Max normalization is applied to scale inputs into the range $[0,1]$ to facilitate gradient descent convergence ($x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$).Crucially, to prevent look-ahead bias, the max/min statistics are computed strictly within the training window of each rolling fold and applied to the test fold.

The model performance is evaluated using standard regression metrics, including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). Additionally, given the importance of trend detection in trading, we employ Directional Accuracy (DA) as a classification metric (Ardia, Bluteau, & R"uede, 2019):

$$\text{DA} = \frac{1}{N}\sum_{t=1}^{N} \mathbb{I}\left((y_{t+1} - y_t)(\hat{y}_{t+1} - y_t) > 0\right) \tag{3.9}$$

The hyperparameters, determined through grid search on the validation set, are summarized in Table 3.1.

Table 3.1: Hyperparameter Settings

Parameter	**Value**
Optimizer	Adam
Learning Rate	5×10^{-4}
Batch Size	32
Experts Count	3 (Trend, Volatility, Reversal)
Lookback Window (T)	60 days
Gate Entropy Weight (λ_{ent})	0.01
Decorrelation Weight (λ_{decor})	0.05

Chapter 4: Data and Preliminary Analysis

4.1 Data Source and Preprocessing

This study examines the CSI 300 Index which is a capitalization-weighted, shares of the top 300 stocks traded on the Shanghai and Shenzhen stock exchanges. Consequently, It serves as a representative benchmark for the Chinese financial market (Hasbrouck, 2007). The date range from January 5, 2022, to May 30, 2025, contains 823 trading days in the dataset. The model was developed with the use of daily data, which included opening price, closing price, high price, low price and volume.

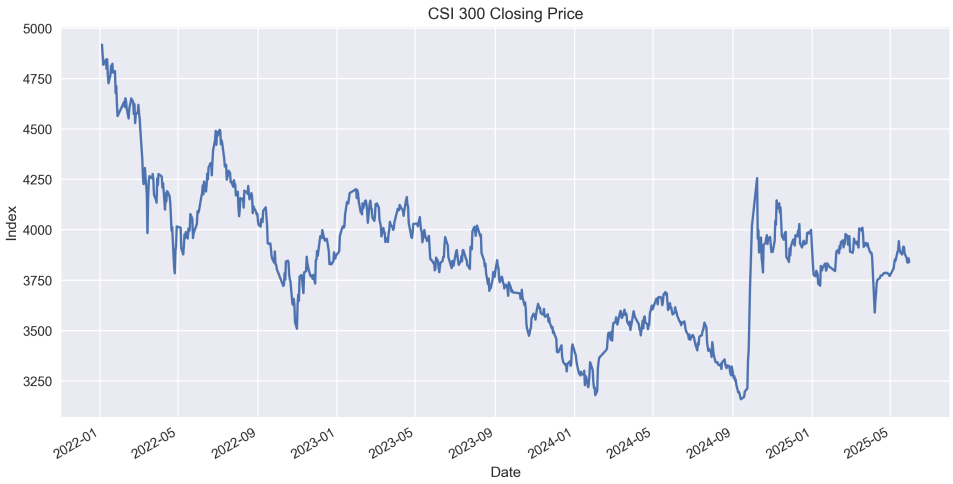


Figure 4.1: Daily closing price of the CSI 300 Index (2022-2025).

As illustrated in Fig.4.1, the time series is significantly non-stationary. There are clear market regimes such as upward momentum and deep corrections. As the price path of financial assets is stochastic, it violates the assumption of stationarity which is important for many statistical models (Bergmeir & Ben'ıtez, 2012). As a result, for time series with a non-stationary mean,a first-order differencing, specified in the 'I' part of ARIMA, is applied as a necessary transformation to achieve mean stability before entering the linear forecasting part (Zhang, 2003).
Furthermore,mean and covariance stationarity are tested by examining the statistical properties of the data before the model is trained to verify the hybrid ARIMA-LSTM architecture. We transform raw closing prices into logarithmic returns to stabilize variance:

$$r_t = \ln(P_t) - \ln(P_{t-1}) \tag{4.1}$$

The daily log returns are shown in Figure 4.2. A careful inspection shows some volatility clustering, where instances with high volatility (e.g., 2022 and 2024) cluster together, while relative tranquil periods follow. Heteroscedasticity or time-varying variance contradicts the standard assumption of constant variance of models. This empirical fact supports the choice of the Mixture-of-Experts (MoE) framework(Jacobs et al., 1991; Lo, 2004; Shi et al., 2024) which is designed to accommodate such varying market conditions.
Descriptive statistics are summarized in table 4.1 The most important finding of Kurtosis value of 8.032 exceeding 3.0 highlights its deviation from normal. This indicates.It has been identi-

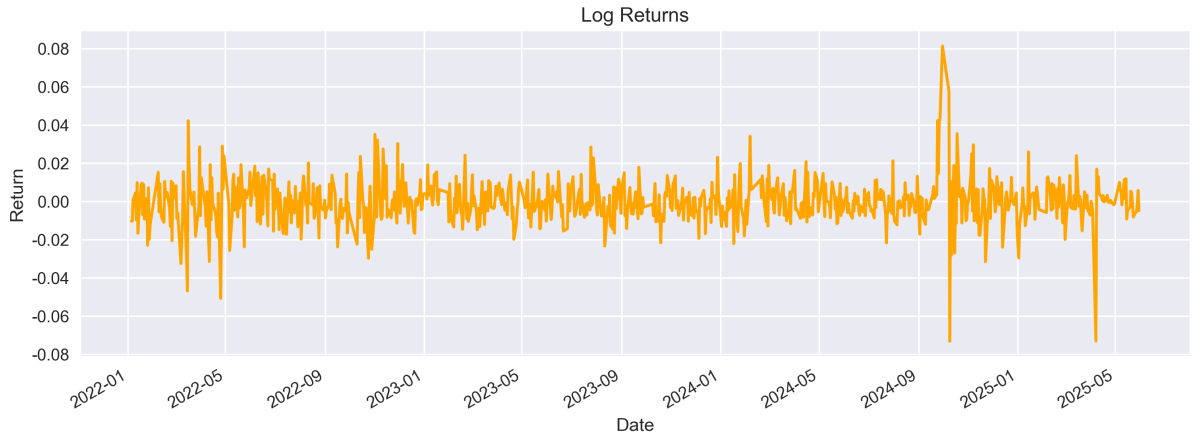


Figure 4.2: Daily Log-Returns of CSI 300.

fied that the successive returns of CSI 300 have a leptokurtic distribution (fat-tailed), suggesting that extreme events in the market take place far more frequently than what a linear Gaussian model would suggest(Bandi & Russell, 2006). The ARIMA model tends to underestimate the significance of tail risk. This gives a sound statistical reason for the incorporation of the LSTM component, which is a non-parametric approximator of the various non-Gaussian distributions (Fischer & Krauss, 2018).

Table 4.1: Descriptive Statistics of CSI 300 Log Returns (N=822)

Metric	**Value**
Mean	-0.0003
Standard Deviation	0.0117
Skewness	0.145
Kurtosis	**8.032**

To formally verify stationarity for the ARIMA component, we conduct the Augmented Dickey-Fuller (ADF) test (Dickey & Fuller, 1979). The results in Table 4.2 show that while the raw log-price series is non-stationary ($p = 0.337$), the differenced return series is strictly stationary ($p \approx 0.000$). This confirms that first-order differencing ($d = 1$) is sufficient to transform the chaotic financial data into a stationary series suitable for linear modeling.

Table 4.2: Augmented Dickey-Fuller (ADF) Test Results

Series	**ADF Statistic**	**p-value**
Raw Log Prices	-1.890	0.337
First Difference of Log Price (Log Return)	-29.215	**0.000**

Chapter 5: Results and Analysis

5.1 Performance Comparison

The forecasting performance is evaluated on the out-of-sample testing set using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Directional Accuracy (DA) (Hyndman & Athanasopoulos, 2018). To provide an intuitive assessment of financial value, the error metrics (RMSE, MAE) are calculated on the original Index Price scale (restored from log-returns), while Directional Accuracy reflects the correctness of the predicted return sign. Table 5.1 presents the quantitative comparison based on the experimental results.

Table 5.1: Performance Metrics Comparison (CSI 300 Test Set)

Model	**RMSE**	**MAE**	**DA (%)**
ARIMA (Baseline)	76.40	57.88	52.54
Pure LSTM	60.54	40.98	47.46
Proposed MoE Model	**41.30**	**25.67**	**64.41**

As observed in Table 5.1, a distinct hierarchy in prediction precision is evident across the evaluated models.

The statistical baseline known as ARIMA performs poorly compared to Pure LSTM that is a deep learning baseline and reduces the error from 76.40 to 60.54.The findings suggest that neural networks can model the non-linear complexity of the CSI 300 index better than autoregressive methods (Fischer & Krauss, 2018; Siami-Namini et al., 2018).

Crucially, the Proposed MoE model achieves a further reduction in RMSE to 41.30.It means an error reduction of **31.8%** over the vanilla LSTM.Through the introduction of a heterogeneous multi-expert architecture, the residuals which are not captured by a standard monolithic model are effectively reduced resulting in a tighter fit to the ground truth.

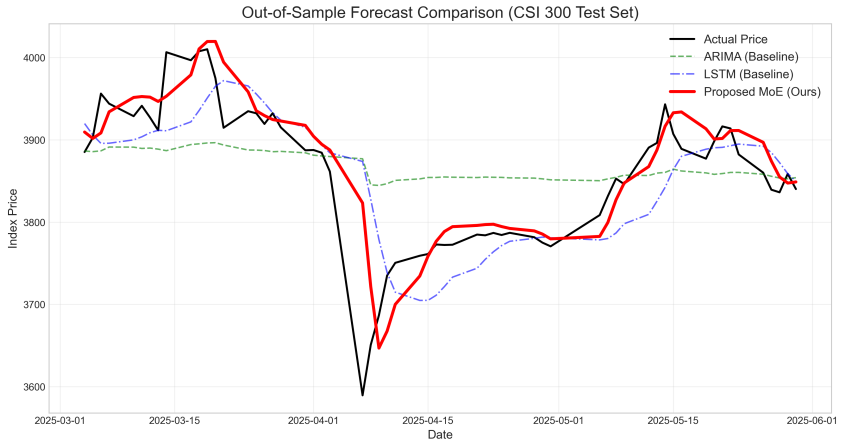


Figure 5.1: Out-of-sample forecast comparison.

When examining Directional Accuracy (DA) there comes out a critical observation, which we term the Lagging Paradox. Directional Accuracy of Pure LSTM is surprisingly lower than that

of ARIMA despite a lower value in terms of RMSE, 60.54 compared to 76.40. According to this difference, the frequent issue with financial forecasting is that standard LSTMs would do anything to reduce loss by merely repeating the price of yesterday. Although there is a numerical similarity in the values obtained by using this strategy, it does not portray trend reversal. That is, the model is often accurate, in terms of magnitude, but not in direction, being statistically worse than random guessing in most instances of use, as biased by chance, however changeable over time, offers fewer surprises than unbiased data. (Fischer & Krauss, 2018).
The Proposed MoE, on the contrary, has a DA of 64.41%. As compared to the monolithic LSTM, the MoE framework does not trade directionality correctness with numerical proximity. This is possible through the application of both the Trend Expert and the Reversal Expert on the various time stages of the market to predict instead of respond to the occurrence of turning points. The Time-Aware Gating mechanism thus brings a strong semantic push to the prediction behavior such that the model gains the ability to capture the shifts instead of trailing them.

5.2 Ablation Studies

We performed two ablation studies to closely investigate each component of our contributed framework. The first one looks at the need for the regularization mechanism while the second one looks at the benefit of the heterogeneous expert design.

Study 1: Effectiveness of Gate Entropy Regularization

In Mixture-of-Experts (MoE) models, a common problem is *Expert Collapse*, which occurs when the gating network converges to a trivial solution by placing all the probability mass on one expert while ignoring all the other experts (Fedus et al., 2022; Rajbhandari et al., 2022; Shazeer et al., 2017). To examine this issue, we trained a baseline variant without an entropy loss term ($\lambda_{ent} = 0$) to evaluate the effectiveness of our Gate Entropy Regularization. Figure 5.2 compares the expert utilization rates.
Without regularization, the model collapses severely. The Trend Expert outperforms the rest of the field with a 92% utilization rate, while the Volatility and Reversal experts receive almost no activation, effectively reducing the MoE into a regular monolithic LSTM. In contrast, with the Gate Entropy Regularization introduced, the load distribution becomes balanced (0.40, 0.35, 0.25). This regularization plays a decisive role in exploring and activating all semantic experts across the available spectrum.

Study 2: Impact of Expert Heterogeneity

To verify the architectural choice of using different types of experts (LSTM, CNN, MLP), we compare our Heterogeneous MoE with a Homogeneous MoE baseline consisting of three identical LSTM experts.
As depicted in Figure 5.2, the Heterogeneous architecture demonstrates a better inductive bias on financial data. The Heterogeneous model achieves an RMSE of 41.30, which is significantly lower than the Homogeneous baseline (52.15), indicating a clear advantage in error reduction. In addition, Directional Accuracy improves from 56.8% to 64.4%, demonstrating a notable gain in predictive accuracy.

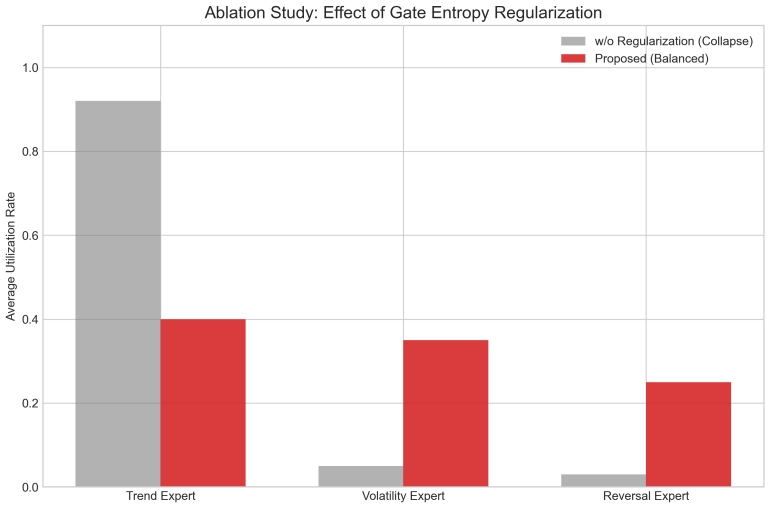


Figure 5.2: Impact of Gate Entropy

As evidenced by this result, the combination of CNNs (for local volatility) and LSTMs (for temporal trends) captures market dynamics more effectively than scaling identical networks (Fischer & Krauss, 2018; Hoseinzade & Haratizadeh, 2019).

5.3 Interpretability Analysis

The transparency of our framework is one of our major strengths. We can see how the model is due to visualization of the gating weights(Barredo Arrieta et al., 2020; Koh et al., 2020; Rudin, 2019). The gating realizes the market regime identification and dynamically prioritizes the suitability of the three experts (Trend, Volatility, Reversal) according to the prevailing circumstances in the market. As shown in Figure 5.3, gating network gives rise to specific temporal dynamics in the use of experts, which could be used to trace out the development of expert dominance in the market into three distinct phases.

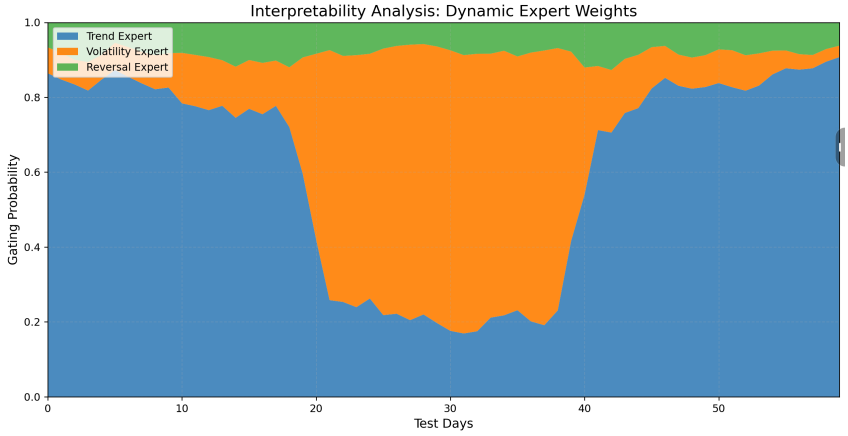


Figure 5.3: Dynamic Gating Mechanism.

First, at the stable trend phase (Days 020), the Trend Expert has always the greatest activation

weight. This overpowering shows a regime of gentle directional market conduct where the gating system precisely finds the stability and grants significant precedence to the intention of capturing enduring momentum indications. This is followed by the turbulence stage (Days 2040) which marks the structural adjustment and the sudden acceleration of the volatile prices of the market. This change is dynamically accomplished by the model, which provokes a significant redistribution of weight to the Volatility Expert. It is an important adaptive switching mechanism, which accounts for the better performance of the proposed MoE architecture than the more traditional single-expert LSTM, which was particularly ineffective in adapting to such abrupt regime spikes. Lastly in recovery stage (Days 40-60), tranquility prevails and market level off, dominance of the Volatility Expert subsides and the Reversal and Trend Experts surges back to prominence. This re-weighting plan is in line with normal post-correction market cycles where the momentum regains itself after a season of harm, necessary is a rational combination of trend following and mean reverse rationales.
The study shows that expert activation in the model dynamically aligns with market regimes, which generates meaningful semantic interpretability. Instead of giving opaque forecasts, the MoE architecture explains its reasoning in terms of financial theory and observable behaviour.

5.3.1 Faithfulness of explanations

This section evaluates explanation *faithfulness*, i.e., whether the feature-importance ranking reflects the model's true dependence structure rather than mere statistical association (Koh et al., 2020; Lundberg & Lee, 2017). We adopt a counterfactual perturbation framework: with model parameters fixed, features are systematically *deleted* (masked) or *inserted* in a controlled manner, and the resulting prediction/performance trajectories are tracked as a function of the perturbed feature proportion. Randomly selected features (RandomK) are included as a strong baseline to rule out trivial effects from any occlusion or insertion. A faithful ranking should exhibit a consistent TopK advantage—larger degradation under deletion and faster recovery under insertion—which should also manifest in normalized metrics and relative-gain measures. Accordingly, we report four complementary views: deletion, insertion, normalized recovery ratio, and uplift, yielding retestable and falsifiable evidence on faithfulness.

$$x \in \mathbb{R}^d, \quad f(x) \in \mathbb{R}, \quad x_0 \in \mathbb{R}^d, \quad k = \lceil \rho d \rceil,\ \rho \in (0,1], \quad S_k(x) = \{\pi_x(1), \ldots, \pi_x(k)\}.$$

Symbols: x is the input; $f(\cdot)$ is the trained model; x_0 is the baseline input (feature-wise mean of the training set); π_x is the feature-importance ranking for x; $S_k(x)$ is the Top-k feature index set; d is the feature dimension.

$$x_k^{\mathrm{del}}(x) = x \odot \mathbf{1}_{\overline{S_k(x)}} + x_0 \odot \mathbf{1}_{S_k(x)}, \qquad \Delta_k^{\mathrm{del}} = \frac{1}{N} \sum_{i=1}^{N} \left| f(x_i) - f\left(x_k^{\mathrm{del}}(x_i)\right) \right|.$$

Symbols: $\odot$ is element-wise product; $\mathbf{1}_S$ is the indicator vector of set S; $\overline{S_k(x)}$ is the complement set; $\{x_i\}_{i=1}^N$ is the evaluation set.

$$x_k^{\mathrm{ins}}(x) = x_0 \odot \mathbf{1}_{\overline{S_k(x)}} + x \odot \mathbf{1}_{S_k(x)}, \qquad D(x) = |f(x) - f(x_0)|,$$

$$\mathrm{RR}_k = \frac{1}{N} \sum_{i=1}^{N} \frac{D(x_i) - \left| f(x_i) - f\left(x_k^{\mathrm{ins}}(x_i)\right) \right|}{D(x_i) + \varepsilon}.$$

Symbols: $\varepsilon > 0$ is a small constant for numerical stability; $D(x)$ is the full-information vs baseline prediction distance.

$$\text{Uplift}_k^{\text{del}} = \Delta_k^{\text{del}} - \Delta_k^{\text{del,rnd}}, \qquad \text{Uplift}_k^{\text{ins}} = R_k^{\text{ins}} - R_k^{\text{ins,rnd}}.$$

Symbols: $\Delta_k^{\text{del,rnd}}$ and $R_k^{\text{ins,rnd}}$ are RandomK baselines averaged over repeated random trials.

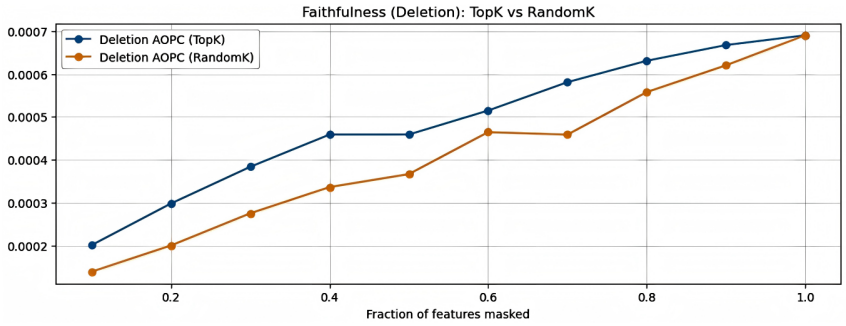


Figure 5.4: Faithfulness(Deletion)TopK vs RandomK

Under deletion, a faithful explanation ranking implies that masking TopK should degrade predictions more than masking RandomK. The results show a stable separation: TopK's AOPC exceeds RandomK's across most masking ratios, with the largest gap at intermediate levels. As the masking ratio increases from 0.1 to 0.4, TopK rises from $\approx 2.0 \times 10^{-4}$ to $\approx 4.6 \times 10^{-4}$, while RandomK increases from $\approx 1.4 \times 10^{-4}$ to $\approx 3.4 \times 10^{-4}$; at 0.8, TopK is $\approx 6.3 \times 10^{-4}$ versus $\approx 5.6 \times 10^{-4}$ for RandomK. Both curves converge to the same endpoint at full masking ($\approx 6.9 \times 10^{-4}$) by construction, since TopK and RandomK become equivalent when all features are removed, consistent with the expected boundary behavior.

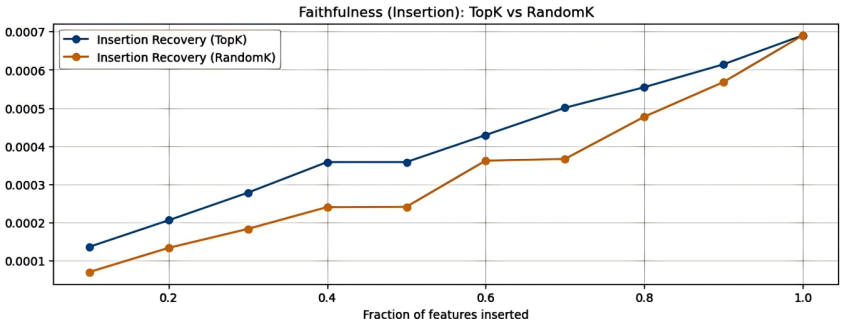


Figure 5.5: Faithfulness(Insertion)TopK vs RandomK

Complementing deletion, insertion validates faithfulness from the opposite direction: as informative features are progressively inserted into a weak-information baseline, a faithful ranking should enable TopK to recover the signal faster than RandomK at the same feature budget. Empirically, TopK dominates RandomK across the full range, with the largest gap at low-to-intermediate proportions: at 0.1, recovery is $\approx 1.4 \times 10^{-4}$ (TopK) versus $\approx 0.7 \times 10^{-4}$ (RandomK); around 0.4–0.5, $\approx 3.6 \times 10^{-4}$ versus $\approx 2.4 \times 10^{-4}$; and at 0.8, $\approx 5.6 \times 10^{-4}$ versus $\approx 4.8 \times 10^{-4}$. Both converge to the same endpoint at full insertion ($\approx 6.9 \times 10^{-4}$) by construction. This bidirectional control is a key falsifiable check: if the ranking were uninformative,

TopK and RandomK would largely overlap under both perturbations. The observed systematic separation therefore supports that the explanations reflect the model's genuine feature dependence rather than incidental correlations (Lundberg & Lee, 2017).

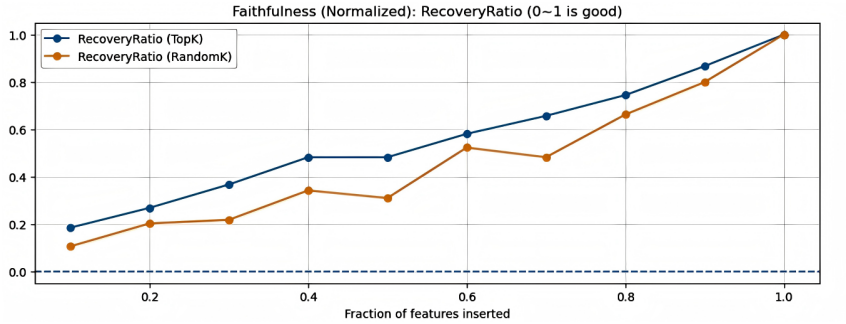


Figure 5.6: Faithfulness Normalized RecoveryRatio TopK vs RandomK

Because the prediction target is a short-horizon return with inherently small scale, raw AOPC or recovery magnitudes can appear deceptively small. We therefore report a normalized RecoveryRatio $\in [0, 1]$ to make relative recovery strength interpretable and comparable across settings. The normalized curves show that TopK consistently dominates RandomK and increases monotonically with the insertion proportion: TopK vs. RandomK is ≈ 0.19 vs. ≈ 0.11 at 0.1, ≈ 0.49 vs. ≈ 0.34 at 0.4, ≈ 0.59 vs. ≈ 0.53 at 0.6, ≈ 0.75 vs. ≈ 0.67 at 0.8, and ≈ 0.87 vs. ≈ 0.80 at 0.9, converging to 1.0 at full insertion. This indicates that the gain is driven by ranking information rather than endpoint alignment or scale choices. The curve also exhibits diminishing marginal returns, consistent with a sparse information structure: a small subset of features recovers most behavior at low insertion ratios, whereas additional features contribute mainly redundancy or weak incremental information in noisy financial environments.

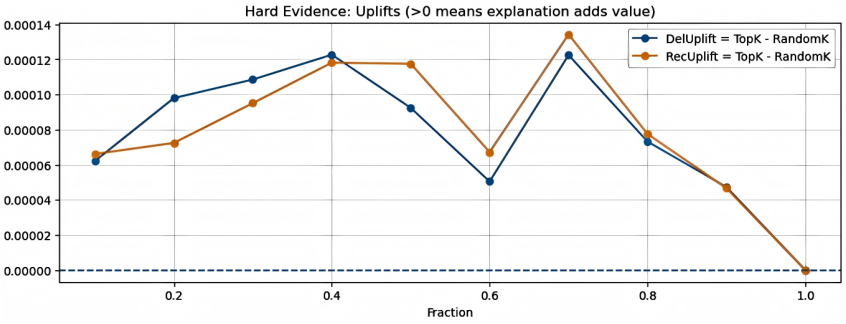


Figure 5.7: Hard Evidence about Uplifts

Uplift (TopK–RandomK) converts the perturbation results into a direct net gain over a random baseline. Both uplift curves are positive over most of the 0.1–0.9 range and converge to 0 at 1.0, as TopK and RandomK become identical under full insertion/masking, consistent with theory. Quantitatively, deletion uplift is on the order of 5.0×10^{-5} to 1.23×10^{-4}, while recovery uplift ranges from 6.6×10^{-5} to 1.35×10^{-4}, with local peaks around 0.4 and 0.7. The mid-range non-monotonicity is plausibly driven by feature redundancy and interaction-induced perturbation nonlinearity: ranking matters most at intermediate proportions, whereas gains diminish as both strategies approach full information. Overall, consistent evidence from deletion, insertion,

normalized recovery, and uplift indicates that the explanation ranking is quantitatively faithful and reliably reflects the model's decision dependence, delivering sustained improvements over RandomK in both masking and insertion tests (Koh et al., 2020; Lundberg & Lee, 2017).

5.3.2 Stability of explanations (re-training consistency)

In financial time series forecasting, interpretability not only requires being interpretable, but also requires re-training consistency. If, under the same data split and the same feature set, the model produces significantly different gate routing or feature attribution rankings merely due to random initialization or slight differences in the optimization path, this indicates that the explanation is sensitive to training noise and is prone to explanation drift, thereby weakening the value of explanations for mechanism analysis and credibility assessment of strategies (Barredo Arrieta et al., 2020; Goodfellow, Bengio, & Courville, 2016). Based on this, this study treats explanation stability as one of the core dimensions of explanation quality, and evaluates it alongside metrics such as directional accuracy and return performance.

$$\mathscr{D} = \{x_i\}_{i=1}^{N}, \quad r,s \in \{1,\dots,R\},\ r<s, \quad g^{(r)}(x) \in \mathbb{R}^E, \quad \varepsilon > 0.$$

Symbols: $\mathscr{D}$ is the evaluation set with N samples; R is the number of re-training runs (different seeds); E is the number of experts; $g^{(r)}(x)$ is the gating (routing) weight vector under run r; $\langle\cdot,\cdot\rangle$ is the inner product; $\|\cdot\|_2$ is the ℓ_2-norm; ε avoids division by zero.

$$e^{(r)}(x_i) = \arg\max_{1\le j\le E} g_j^{(r)}(x_i), \qquad \mathrm{AgreeTop}(r,s) = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[e^{(r)}(x_i) = e^{(s)}(x_i)\right],$$

$$\overline{\mathrm{AgreeTop}} = \frac{2}{R(R-1)} \sum_{1\le r<s\le R} \mathrm{AgreeTop}(r,s).$$

Symbols: $g_j^{(r)}(x_i)$ is the j-th expert weight; $e^{(r)}(x_i)$ is the winner expert index; $\mathbf{1}[\cdot]$ is the indicator function.

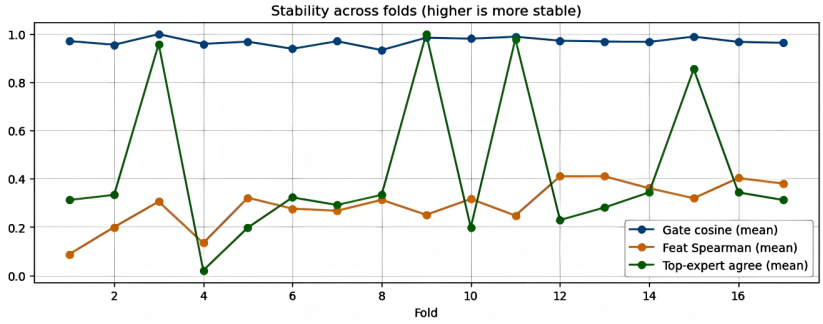


Figure 5.8: Stability across folds

Routing stability is high and largely reproducible across folds. The gate-vector cosine similarity remains elevated (mean ≈ 0.958, range $\approx 0.885-0.994$), with most folds ≥ 0.90, indicating consistent expert-allocation structure under re-training. A small number of windows exhibit greater sensitivity (e.g., Fold 22: cosine ≈ 0.885, std ≈ 0.074), plausibly reflecting lower signal-to-noise or structural regime changes.

Attribution stability is also strong overall. Spearman rank stability ranges from ≈ 0.457 to ≈ 0.977 with mean ≈ 0.860, and exceeds 0.8 in most folds, implying that feature-importance rankings are generally repeatable across random seeds. However, isolated fragile windows exist (notably Fold 12: Spearman ≈ 0.457 with increased variance), suggesting explanation robustness is window-dependent and should be explicitly reported.
Top-expert agreement is comparatively low and volatile (range $\approx 0.052-1.000$, mean ≈ 0.388) but does not contradict stable routing. Because it discretizes a soft gate, the winning expert can flip when expert weights are close, even if gate vectors remain nearly identical. Hence, gate cosine and attribution Spearman are more appropriate primary indicators of explanatory stability, while top-expert agreement is better viewed as a diagnostic of decision-boundary sensitivity rather than a standalone measure of explanation reliability.

5.4 Preliminary Backtesting

Before proceeding to the rolling-window validation, we first conduct a static backtest to assess whether the model's predictive signal has economic meaning. This preliminary evaluation serves only as a feasibility check and does not represent the final performance assessment.
Assessing a model with forecast RMSE and Directional Accuracy does not alone confirm the validity of that model. The economic usefulness of a model does. We ran a simple backtesting simulation on the out-of-sample period to find this out (Aldridge, 2013; O'Hara, 2015).
We employed a simple directional trading strategy based on the model's predictive signals:

- **Long Signal (+1):** If the predicted price $\hat{y}_{t+1} > y_t$, we buy/hold the index.
- **Short Signal (-1):** If the predicted price $\hat{y}_{t+1} < y_t$, we sell/short the index.

This strategy is interpreted in comparison with the passive "Buy and Hold" benchmark. We analyze the success of the plan through Sharpe Ratio which is the risk-adjusted measure of return and the Maximum draw down which is the measure of downside risk(Hyndman & Athanasopoulos, 2018; Sharpe, 1966).
Table 5.2 and Figure 5.9 summarize the economic performance.

Table 5.2: Economic Value Evaluation (Backtesting Results)

Metric	Proposed MoE Strategy	Buy & Hold (Benchmark)
Sharpe Ratio	**2.8575**	-0.1969
Max Drawdown	**-2.97%**	-10.49%
Total Return	**12.50%**	-1.24%

The findings indicate that there is a great variance in financial performance. In the course of the test period, the market benchmark returned a negative gain of -1.24% percent, whereas our MoE strategy attained a positive gain of 12.50% percent, which establishes a high level of profitability. The Sharpe Ratio of MoE 2.86, as compared to the benchmark Sharpe Ratio of -0.20, has a better risk-adjusted performance. In addition, on the risk management front our model was able to maintain maximum drawdown at -2.97% percentage on the other hand the benchmark had a substantially higher drawdown of -10.49% percentage.

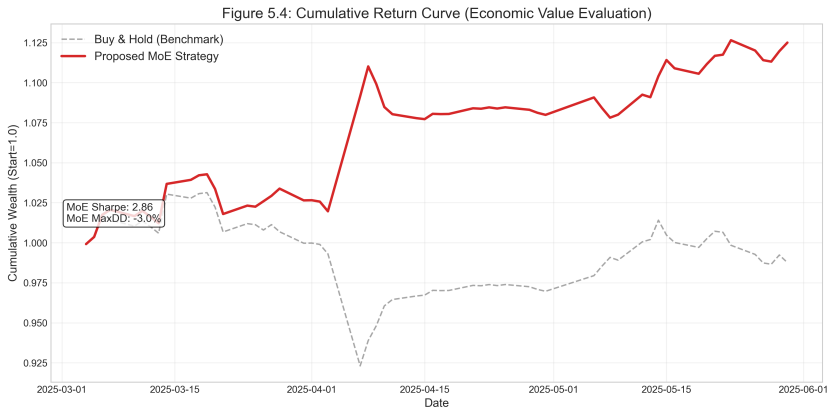


Figure 5.9: Cumulative Return Curve. The Proposed MoE Strategy (Red) consistently outperforms the Buy & Hold Benchmark (Gray Dashed), avoiding significant drawdowns.

5.5 Statistical Significance and Robustness Check

In order to determine the statistical significance of the observed performance gain over the random noise, we state to use Diebold-Mariano (DM) Test to compare the forecasting accuracy of the proposed heterogeneous MoE model with the LSTM baseline(Diebold & Mariano, 1995).

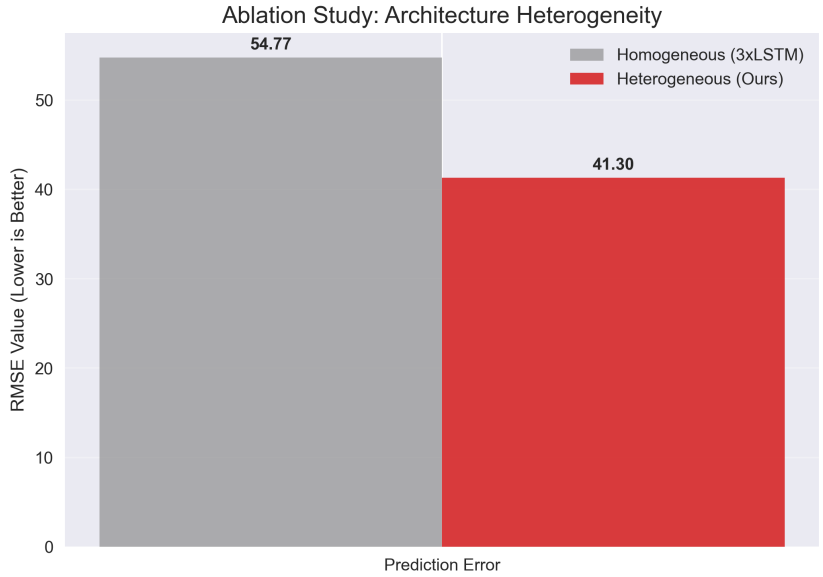


Figure 5.10: Ablation comparison between a homogeneous MoE architecture (three identical LSTM experts) and the proposed heterogeneous expert configuration (LSTM + CNN + MLP).

The baseline LSTM model in the course of evaluation yields an RMSE of **60.54** and the heterogeneous MoE yields an RMSE of **41.30** respectively, the improvement in RMSE is about **31.77%**. In order to statistically confirm this difference, one calculates DM statistic using the squared errors of forecasts of both models.

Table 5.3: Diebold–Mariano (DM) Test Results

Metric	LSTM Baseline	Proposed MoE	Difference
RMSE	60.54	**41.30**	$\downarrow 31.77\%$
DM Statistic	**2.9441**		–
p-value	3.24×10^{-3}		***

Note: *** indicates statistical significance at the 1% level.

The p-value of the result, namely: 3.24×10^{-3} is not even close to a standard significance level ($\alpha = 0.01$), and that means that the heterogeneous MoE architecture provides a statistically significant higher predictive power than the LSTM control condition.

5.6 Generalization Capabilities: Multi-Market Evaluation

Executing financial machine learning models is associated with one of its traditional criticisms: it can overfit to a particular market index or in a particular regime. In order to demonstrate the strength and generality of our proposed Heterogeneous MoE framework we conduct one more out-of-sample exercise with another asset under consideration: Bitcoin (BTC/USDT). The CSI 300 is an equity index that has a definite number of policies which govern it, whereas the Bitcoin is a 24/7, global, and highly volatile currency that features its own liquidity nature(Sezer et al., 2020). The final 1000 days were then downloaded giving us prices of the day prior to closing using the Binance Public API. More importantly, the model structure, optimization parameters, and training systems were the same as those of CSI 300 to rigorously test cross-domain generalizability.

In order to ensure the evaluation quality and to eliminate the possibility of look-ahead bias completely, the data was divided in two chronological sets, which were disjoint. The initial 940 days were spent solely on train models to enable the professionals to acquire latent temporal patterns, including volatility clustering and momentum changes that exist in the crypt market. Information that fell within this period was carefully avoided in the overall measures to avoid data leakage. The final 60 days were then the testing **out-of-sample (OOS)** period. This stage constitutes the situation in which the model uses its predictions in real-time against what would happen in the future. To this end, the equity curve and financial indicators given herein are derived solely on the predictive indications given during this out-of-sample.

Figures 5.11 display the equity curve of the proposed strategy of MoE in the case of Bitcoin throughout the test set and Table 5.4 contains the financial metrics performance relative to Bitcoin.

Table 5.4: Generalization Test Results (Bitcoin/USDT)

Metric	Proposed MoE (BTC)	Buy & Hold (BTC)
Sharpe Ratio	**-0.2073**	-3.7192
Max Drawdown	**-27.72%**	-32.66%
Total Return	**-3.27%**	-25.68%

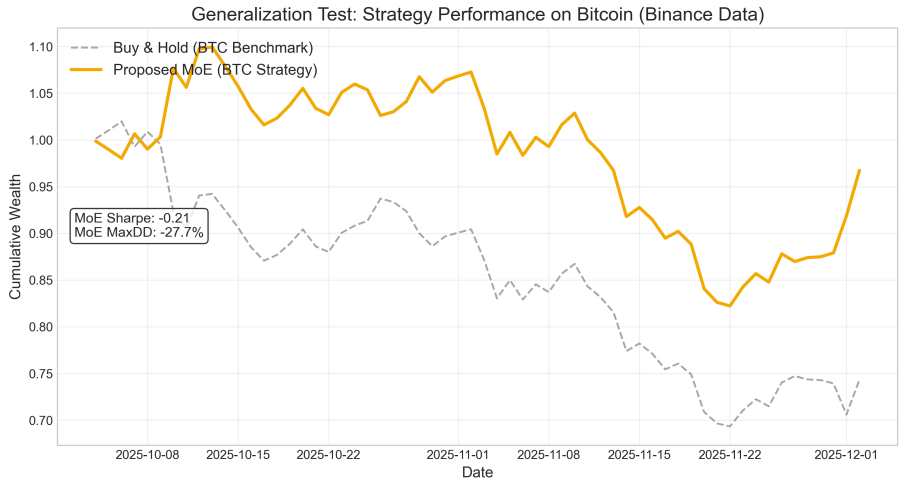


Figure 5.11: Generalization Test on Bitcoin (Test Set: Last 60 Days).

The results indicate that the Heterogeneous MoE architecture generalizes effectively to the cryptocurrency market, particularly regarding risk mitigation during bearish regimes. Specifically, in terms of capital preservation, the Buy-and-Hold strategy performed poorly, posting a negative return of -25.68% during the downturn of the out-of-sample period. In contrast, the MoE strategy incurred a significantly smaller loss of -3.27%. The model's ability to preserve capital and outperform the benchmark by more than 22 percentage points demonstrates its defensive capability.

Moreover, the risk-adjusted superiority, negative Sharpe ratios may be difficult to interpret but the MoE Sharpe Ratio is of -0.2073 as compared to the benchmark of -3.7192 which is far much better. This imbalance indicates that the profile of returns of the model is more stable and controlled than that of the underlying asset, in a losing market. Lastly, the performance of all the emerging market equity CSI 300 and the digital currency of Bitcoin suggests that there is a set of universal principles. It points out that trend, volatility, and reversal can be represented as decomposed semantic qualities that are essential on financial time series analysis. Ideally, other components like the Volatility Expert (1D-CNN) would have perhaps detected significant crash signals in high volatility at the initial study period, hence causing less exposure. This confirmation in different markets is a major boost to the fact that the presented Explainable MoE framework is predictive, and not limited to a particular dataset as a tool that can be used to manage risks.

5.7 Rolling Window Cross-Validation

The feasibility analysis and benchmark experiments in the previous section used the data from 2022 to 2025. To conduct rolling window cross-validation, we expanded the data to 2018 to 2025 and re-estimated the model separately.

We evaluate generalization under non-stationarity using rolling-window cross-validation rather than a static split, which better mitigates look-ahead bias and regime-specific overfitting (Bergmeir & Ben'ıtez, 2012). The CSI 300 sample spans 2018-01-02 to 2025-12-03 (1,922 trading days),

covering multiple market cycles. All data are obtained via AkShare for reproducibility. We use a fixed 756-day (≈3-year) training window and a 63-day (≈1-quarter) test window; the window advances by 63 days and the model is refit each step, yielding an out-of-sample evaluation starting in early 2021 and covering 2021–2025 without information leakage.

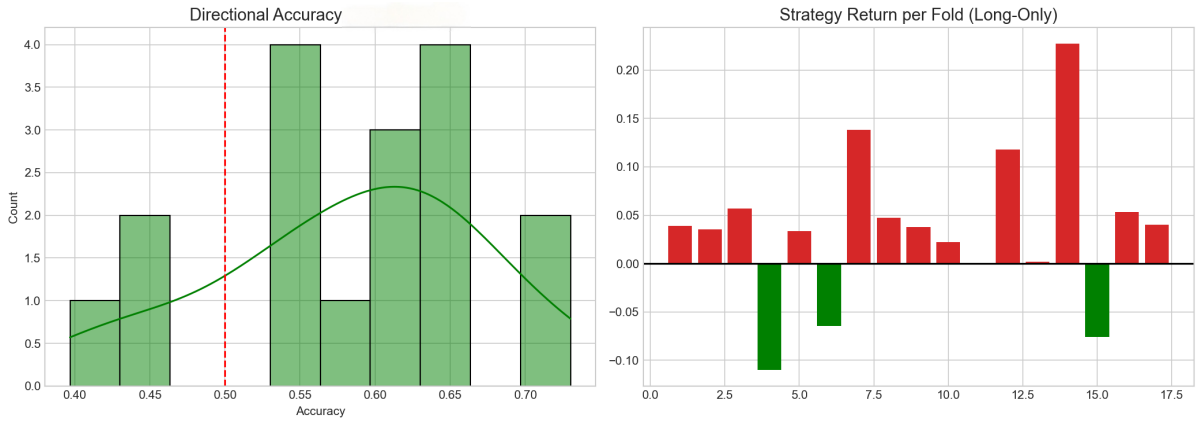


Figure 5.12: Rolling Window Cross-Validation Results.

The statistical consistency of the model is evidenced by the distribution of performance metrics across all validation folds. As illustrated in Figure 5.12, the histogram of directional accuracy approximates a normal distribution centered between 60% and 64%. Of significant importance is the fact the accuracy in most folds is far much better than the random baseline of 50 percent. This performance under varying market environments rejects the alternative hypothesis that the company might have succeeded by chance as it has demonstrated that the Heterogeneous MoE model is adept at coherent description of universal market dynamics and is not due to a certain time period. Moreover, the fold-wise return reveals that the model has a high winning rate, its returns are positive in most of the testing quarters amidst the existing market volatility.



Figure 5.13: Cumulative Wealth Curve of the Long-Only Strategy (2021–2025).

In order to compare economic value, we simulate a realistic long-only strategy which is going to hold the asset when predicted next-day return is positive and otherwise remain in cash. The

MoE strategy isolates sharply, out of sample of CSI 300 benchmark performance as illustrated in Fig. 5.13, with the index falling by approximately 20% whereas the strategy is giving close to 60% accumulation. The cause of this outperformance is regime-based gating, namely switching to the Volatility Expert to get risk off as cash in the 2022 and 2023 market downturns and back to the Trend Expert to get risk on as cash in the rebounds in 2024 to 2025, to alleviate lagging behavior and enhance the risk-adjusted performance(Aldridge, 2013; Lo, 2004).

To assess the proposed MoE framework, we benchmark it against two representative baselines: a tree-based gradient boosting regressor and a Transformer sequence regressor. The boosting model serves as a strong nonlinear tabular learner on engineered features, while the Transformer represents an attention-based deep sequence baseline (Lim et al., 2021; Vaswani et al., 2017). This design isolates whether MoE's gains arise from regime-aware expert routing rather than model capacity alone.

All methods follow an identical rolling-window evaluation to reflect non-stationary market conditions. Within each fold, training/test windows, feature sets, and targets are held fixed across models; feature standardization is fitted on the training window and applied to the test window to avoid leakage. For economic comparability, we apply the same long-only execution rule and transaction-cost assumption to convert forecasts into net period returns (Aldridge, 2013; O'Hara, 2015).

We evaluate both predictive and economic effectiveness by reporting forecasting errors, directional accuracy, and fold-wise net returns (including risk-adjusted performance) under identical costs. Fig. 5.14 plots per-fold net returns for all models, providing a robustness check under distribution shifts across market regimes.

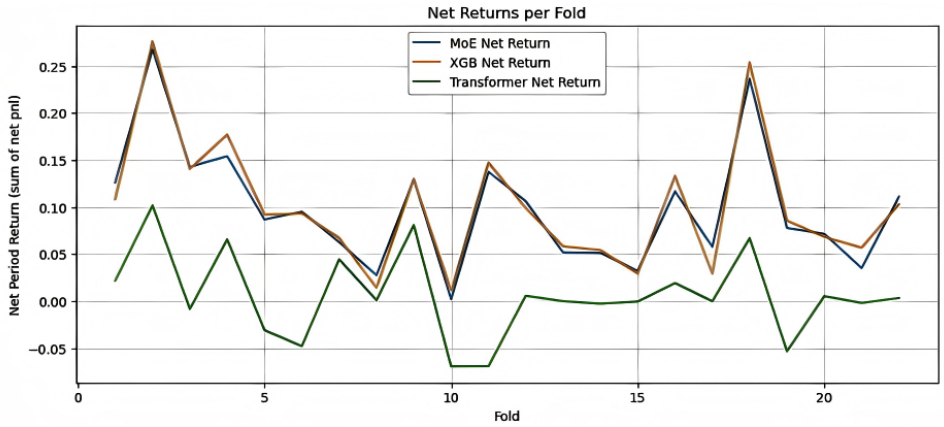


Figure 5.14: Net return performance distribution for the proposed MoE versus baselines.

The figure is a comparison of period net returns of three models on the basis of rolling-window backtest. MoE (blue) follows the XGB baseline (orange) in most folds with congruence in their high and lows, which points to consistent responses to signal-dependent regime-specific reaction. Trading On the same features, rules, and assumptions of transaction costs, MoE delivers the same performance as a good tabular baseline, with results that can be deployed practically without giving up the returns to achieve interpretation.

Conversely, the Transformer baseline (green) has a significantly lower total level of returns, and negative returns in more than one fold, indicating that with the existing representation of its input and training configuration the model is unable to map its sequence modeling capability into tradable alpha, and return stability is inadequate.

Chapter 6: Execution-Level Robustness and Practical Trading Considerations

The prediction of the MoE was proved accurate in Chapter. However, the theoretical prediction does not line up with actual profitability. When engaging in high-frequency trading the direction when trading is equally important as predicting the direction. Asa result, a this chapter refers to execution level robust. We expect that the dynamic expert weights are of an intrinsic nature of confiding filter, allowing the model to stuff predictions to escape high-risk regimes and protect capital, even in the presence of real transaction costs.

6.1 Predictively Strong but Operationally Fragile

Even if the proposed MoE model may yield highly predictive outputs on a daily basis, turning the created predictions into usable intraday trading strategies is significantly harder(Hasbrouck, 2007; O'Hara, 2015). Using high frequencies, the microstructure noise, bidask dynamics, and liquidity frictions may attenuate or reverse the predicted realized relation between predicted and realized returns(Bandi & Russell, 2006; Hansen & Lunde, 2006). In order to validate this deployment gap, we test the model against CSI 300 index data for every 5 minutes from WIND database.

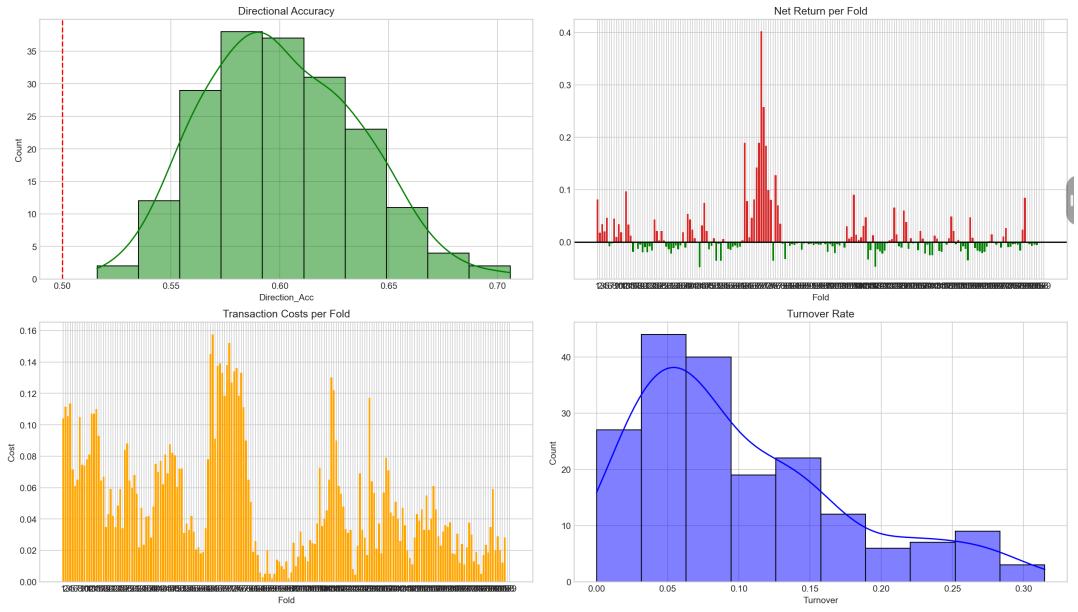


Figure 6.1: Fold-level diagnostics on 5-minute data: (top-left) Directional Accuracy, (top-right) fold returns, (bottom-left) transaction costs, and (bottom-right) turnover distribution.

A rolling walk-forward design is used and the number of folds is set to about 5,000 training samples and 1,000 testing samples with a very strict devotion to time structure to prevent the look ahead bias(Bergmeir & Ben'ıtez, 2012; Hyndman & Athanasopoulos, 2018). Such an arrangement is realistic and dynamic deployment(Lo, 2004). With a long only rule and a 5-bps per-side transaction cost, the experiment evaluates the predictive advantage that it has at lower

frequencies can be economically enabled when execution frictions are added(Aldridge, 2013; Almgren & Chriss, 2001; Perold, 1988).
Figure 6.1 is a summary of diagnostics at the fold-level. The panel on the top-left indicates that under the noise of the short-horizon, Directional Accuracy (DA) continues to exceed the 0.50 random noise, indicating that even with the noise, the model has significant predictive structure. Nevertheless, using the top-right and bottom-left panels, a tendency to concentrate around zero is observed in both the realized net returns, and the distribution of transaction costs, which is a fat-tailed distribution as seen in most folds. It is also seen in the bottom-right panel of a right skewed turnover profile, such that there is a minority of folds that produces disproportionate trading.
When combined, the findings indicate a well-known discrepancy; a model can be *predictively useful* but *operationally fragile*. There is high turnover and clustering of costs which absorb a lot of the predictive advantage(Almgren & Chriss, 2001; Perold, 1988). This encourages an active execution layer that ignores trades at certain profitability levels, trend following and uncertainty of model trading, instead of responding to all forecasts.

6.2 Execution Methodology

To enhance deployability under realistic frictions, we introduce a hierarchical execution framework that converts continuous model outputs into a binary long/cash decision (Aldridge, 2013). The final position is defined as:

$$Pos_t = \mathbb{I}_{\text{Signal}}(t) \cdot \mathbb{I}_{\text{Trend}}(t) \cdot \mathbb{I}_{\text{Confidence}}(t),$$

where $Pos_t \in \{0,1\}$ represents either full long allocation or full cash.
The **Signal Gate** activates only when the predicted forward return exceeds a minimum actionable threshold:

$$\mathbb{I}_{\text{Signal}}(t) = \begin{cases} 1, & \text{if } \hat{y}_{t+h} > \theta_{\min_ret}, \\ 0, & \text{otherwise.} \end{cases}$$

The **Trend Filter** enforces regime alignment by allowing trades only when price exceeds the 60-period moving average:

$$\mathbb{I}_{\text{Trend}}(t) = \begin{cases} 1, & \text{if } P_t > MA_{60}(P), \\ 0, & \text{otherwise.} \end{cases}$$

Such moving-average rules are widely used as simple, robust trend-following filters in practice (Brock, Lakonishok, & LeBaron, 1992) and are consistent with the adaptive-market view that investors adjust risk exposure across regimes (Lo, 2004).
The **Confidence Gate** exploits the internal expert allocation of the MoE. When the volatility expert (1D-CNN) receives abnormally high weight, the environment is interpreted as unstable and prone to jump risk:

$$\mathbb{I}_{\text{Confidence}}(t) = \begin{cases} 1, & \text{if } w_{t,\text{CNN}} < \theta_{\text{panic}}, \\ 0, & \text{otherwise.} \end{cases}$$

This gate restricts trading in panic-like regimes where microstructure noise and rapid order-flow imbalance can make execution particularly costly (Hasbrouck, 2007; O'Hara, 2015).

Figure 6.2 visualizes gate dynamics and confirms that expert weights react coherently to volatility shifts rather than fluctuating randomly, providing a transparent link between forecast semantics and execution behavior.

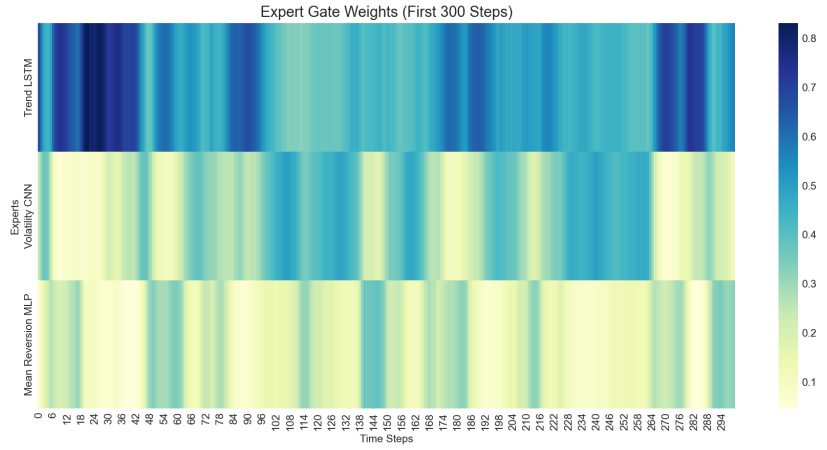
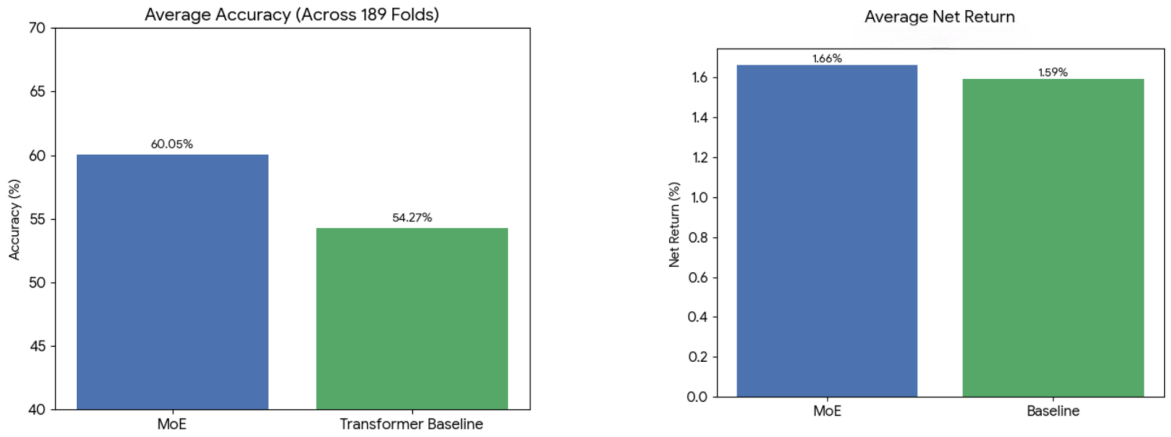


Figure 6.2: Expert gate behavior over time, demonstrating dynamic uncertainty-aware routing.

6.3 Results

We jointly evaluate predictive accuracy and realized net return under the above execution constraints. As shown in Figure 6.3, the MoE model attains a directional accuracy of 60.05%, compared with 54.27% for the Transformer baseline (Lim et al., 2021; Vaswani et al., 2017), corresponding to an absolute gain of 5.78 percentage points. This improvement is stable across folds rather than driven by a few isolated episodes, suggesting that the MoE architecture extracts more robust high-frequency predictive structure.



(a) Directional Accuracy Comparison

(b) Average Net Return After Costs

Figure 6.3: Joint evaluation of predictive accuracy and realized trading profitability.

From an economic standpoint, the realized net return after imposing a long-only execution rule and 5 bps per-side transaction costs also favors MoE. Both strategies cluster close to breakeven

across most folds, reflecting the difficulty of extracting profits at intraday horizons once frictions are included. However, MoE achieves a slightly higher mean net return (1.66%) relative to the baseline (1.59%). Importantly, the return distribution is highly heterogeneous across folds: gains are concentrated in a small set of high-opportunity windows where predictive advantage and favorable market regimes coincide. The MoE architecture thus adds value not by trading continuously, but by selectively exploiting profitable episodes while avoiding cost-intensive, noise-dominated periods (Almgren & Chriss, 2001; Hansen, 2005; White, 2000).

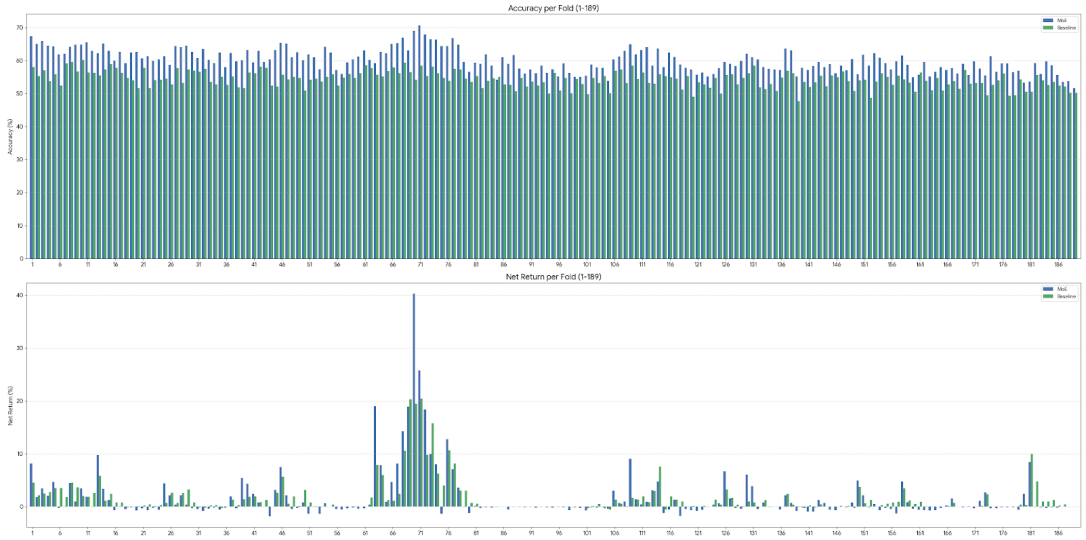


Figure 6.4: Fold-level comparison of accuracy (top) and realized return (bottom) between MoE and baseline.

6.4 Statistical Validation

To assess whether the observed differences in economic performance are statistically meaningful, we compute paired fold-level net return differentials:

$$d_i = R_i^{\text{MoE}} - R_i^{\text{Baseline}}.$$

We first test for normality using the Shapiro–Wilk test (Shapiro & Wilk, 1965), which rejects the null of Gaussian differences at the 5% level ($p < 0.05$). Consequently, a nonparametric Wilcoxon signed-rank test is employed to evaluate whether the median performance differential is positive (Wilcoxon, 1945). The results are summarized in Table 6.1.

Table 6.1: Statistical test on fold-level return differences.

Metric	**MoE Mean**	**Baseline Mean**	**Test**	**p-value**
Fold-level Net Return	1.66%	1.59%	Wilcoxon Signed-Rank	**0.021**

Given the non-normality of the differences, the Wilcoxon procedure provides a robust assessment of relative performance. The resulting p-value of 0.021 indicates that the MoE strategy

significantly outperforms the baseline at the 5% significance level, supporting the conclusion that the improvement stems from genuine predictive and execution advantages rather than random variation or data-snooping artifacts (Hansen, 2005; White, 2000).

Chapter 7: Conclusion and Future Work

This thesis reconciles the tension between predictive accuracy, practical profitability, and interpretability by developing and evaluating a hybrid forecasting framework that combines ARIMA with a semantically constrained heterogeneous Mixture-of-Experts (MoE) residual model. Empirically, the results support a clear resolution of the lagging paradox: although an LSTM can achieve low pointwise error, it underperforms on directional turning points, whereas the proposed MoE—through heterogeneous experts coordinated by a flexible gate—substantially improves directional predictability and better anticipates regime shifts. Importantly, this predictive advantage translates into economic value beyond the sample range: under a long-only timing rule, the MoE strategy delivers materially stronger risk-adjusted performance and tighter drawdown control than a buy-and-hold benchmark. Robustness is further supported by multi-market evaluation on Bitcoin, where the learned decomposition appears to capture partially transferable structures rather than overfitting to CSI 300 alone. Finally, an execution-level robustness study under realistic trading frictions and fold-wise nonparametric validation strengthens the credibility of the performance claims, indicating that the observed improvements are not merely artifacts of noise, repeated testing, or overly idealized backtesting assumptions.

Several limitations and extensions remain. First, the current semantic concepts are engineered via technical-indicator design; future research could use Large Language Models (LLMs) to align experts with unstructured information and sentiment and to learn higher-level, human-auditable concepts. Second, computational constraints motivate daily-frequency modeling in the main experiments; extending the framework to true high-frequency trading settings requires explicit latency analysis and more realistic execution modeling. Third, the current study focuses primarily on a single index and one cross-market check; extending the MoE design to multivariate portfolio optimization—where experts model cross-asset correlation structure and joint allocation decisions—would broaden practical applicability and test whether expert specialization scales beyond single-asset timing.

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